

Bias in LLM-Based Resume Screening Systems

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Abstract

Hiring processes have changed a lot over the years and are now increasingly using AI in processes. AI is often used to screen and shortlist resumes, and this is raising concerns about there potentially being bias in this form of decision-making. The main goal of this study is to look at whether these AI-based screening systems are producing unequal outcomes across groups of different demographics. Many synthetic resumes were created with controlled variations in demographic indicators, including names that were associated with certain genders or ethnicities. Then, these resumes were processed through different LLM-based resume screening systems to analyze the difference in scores based on a controlled rubric system. To address issues, a fairness-oriented tool was made to be used by small businesses and companies to screen resumes with more speed while also being equitable and improving fairness in hiring decision. This project shows how AI systems can be reinforcing existing inequalities that are present in society if they are not designed to look out for them. The findings and process emphasize the importance of having fairness and transparency in automated decision-making systems and show how blind trust can lead to less equitable recruitment practices.

Introduction

Notice the Introduction is a SECTION. Sections can have SUBSECTIONS (seen later)

Open with the broad problem area and narrow toward your specific research question. Why should the reader care? Ground the motivation in real-world impact or a gap in the literature.

Discuss existing solutions and their shortcomings. For example, Smith et al. demonstrated that convolutional approaches plateau at roughly 87% accuracy on this task [1], while the transformer-based method of Jones and Lee ? improved recall but at significant computational cost.

1.1 Background Artificial Intelligence systems are becoming increasingly utilized in modern hiring and recruitment processes. This is especially true during the early stages of resume screening and candidate evaluation. Many companies use automated applicant tracking systems (ATS) and AI-assisted screening tools. This is due to the fact that many companies are receiving very large volumes of applications and this makes them much more efficient. Recently there have been more advances in large language models (LLMs) have made the role of AI in hiring even more extensive as it is now able to be used to summarize, rank, and evaluate resumes according to job descriptions and predefined criteria.

These systems have the capability to influence which applicants receive interviews and employment opportunities, and this has raised concerns regarding their fairness and demographic bias in any automated hiring decisions. Small biases in AI screening systems can scale, especially if the system is looking at thousands of applicants. This can affect access to employment and other opportunities on a much larger scale. There have been many real-world cases of this occurring. For example, Amazon’s experimental AI recruiting tool was trained on historical hiring data, and it unintentionally learned and reproduced these existing societal patterns as it went on to discriminate against women’s colleges and other things that would only be present on women’s resumes.

Also, it has been found that language models and other AI systems may reflect demographic associations that are related to socioeconomic background, gender, and ethnicity. Prior studies have documented bias in many of AI based systems. These include facial recognition, automated decision systems used to financial loan approval, recommendation systems, and natural language processing [2], [3]. Hiring workflows are using LLMs more and more and this has made evaluating the fairness and reliability of these systems a very important area to research and understand.

This study investigates whether demographic signals that are reflected with gender and ethnicity influence the scoring outputs of LLM-based resume screening systems. Through the control of resume content and modifying demographic indicators such as names, this experimental process has a goal to isolate and analyze potential differences in automated resume evaluation outcomes.

1.2 Existing Research Recent research has increasingly looked into the presence of demographic bias in artificial intelligence systems. Research has been particularly in applications involving language models, recommendation systems, and automated decision-making tools. Studies have shown that large language models tend to reproduce patterns and stereotypes present in their training data, including associations related to gender, race, and socioeconomic background [2], [3]. Essentially, this is how they work, by looking at very large sections of data and looking for patterns to use as a basis when making decisions on their own. Researchers have also investigated the use of AI systems in hiring and resume evaluation workflows. Prior studies examining AI-assisted hiring tools have identified concerns regarding fairness, transparency, and the potential influence of demographic information on automated candidate evaluations [4]. In some

cases, machine learning systems trained on historical hiring data were found to unintentionally favor patterns associated with previously successful candidates, potentially reinforcing existing inequalities in hiring practices. More recent work has specifically explored demographic bias in LLM-based resume screening systems. Several studies have tested whether resumes containing different gender- or race-associated signals receive different evaluations despite having similar or even the same qualifications [1], [5]. Other research has examined subtle demographic indicators and their effects on ranking, recommendation, and candidate selection outcomes [1]. These studies suggest that even indirect demographic signals can have an influence in model behavior under certain conditions. In addition to identifying bias, researchers have proposed various methods for evaluating fairness in AI systems, including controlled input testing, comparative scoring analysis, and statistical significance testing [1], [3]. The main goal of these approaches is to isolate demographic variables while minimizing unrelated differences between test cases. These methodologies have become increasingly important as organizations continue integrating AI-assisted tools in high-impact decision-making environments.

1.3 Research Gaps

Although previous studies have explored bias in AI-assisted hiring systems and large language models, there are several limitations in this existing research. Many studies evaluate hiring bias using complex real-world datasets that contain multiple changing variables simultaneously. Most of the time, these include differences in education, experience, formatting, writing style, and skill sets. If many resume characteristics are changing all at once, it can be difficult to isolate whether demographic signals alone influence model behavior. In addition, some past research had looked primarily at overall hiring outcomes or ranking behavior rather than detailed scoring differences across controlled resume inputs. While several studies have investigated demographic bias in language models, fewer studies go in depth at specifically examining LLM-based resume scoring systems. Also these experiments and studies focus on isolating changes in only demographic indicators, which makes it difficult to tell exactly what caused changes in scoring. As a result, the direct influence of gender- and ethnicity-associated signals on automated resume evaluation remains a very much active area of research. This study attempts to address parts of this gap by using a controlled experimental design in which resume content, qualifications, and job descriptions are fixed and only the demographic indicators such as names are modified. By minimizing unrelated variables, this methodology has the main intent to better isolate potential differences in scoring behavior associated with demographic signals. In addition, this project combines controlled experimentation with automated large-scale testing and statistical comparison methods to support more reproducible evaluation of LLM-based hiring systems.

1.4 Research Question

To what extent do gender- and ethnicity-related demographic signals affect the scoring outcomes of LLM-based resume screening systems?

This study investigates whether demographic signals associated with gender and ethnicity influence the scoring behaviors and outcomes of LLM-based resume screening systems. To examine this question, a controlled experimental design was developed in which resume content remained fixed and the only thing that was changing between test groups were demographic indicators.

This project contributes a controlled evaluation framework for analyzing demographic sensitivity in LLM-based hiring systems, along with an automated workflow designed to support scalable and reproducible future experimentation.

Procedure/Experimental Design

2.1 Resume Dataset Design

To evaluate whether demographic signals influence LLM-based resume scoring systems, a controlled resume dataset was developed for experimentation. All resumes used identical qualifications, skills, experience, formatting, and project descriptions. The only modified variable across all of the test cases was the applicant name, which served as the demographic indicator for gender- and ethnicity-associated testing groups. A fixed job description was also maintained throughout each experimental round to ensure consistency in the evaluation criteria. By controlling resume content and minimizing any and all unrelated variation, the experimental design aimed to isolate potential differences in scoring behavior associated with demographic signals rather than applicant qualifications. The goal was to see how the scoring changed based on just applicant names, not based on qualifications. The resumes were generated using a standardized template to maintain consistency across formatting, structure, and wording. This approach reduced the likelihood that variations in resume writing style or formatting would have an influence on the scoring outputs produced by the language model system, as that is not the intention.

2.2 Demographic Groups

The experimental testing was divided into demographic groups associated with both gender and ethnicity. For the gender-based experiments, resumes were generated using male-associated, female-associated, and gender-neutral names while maintaining identical resume content and qualifications. These name associations were made on the general usage of these names, according to multiple baby naming websites. Initial testing was conducted using smaller sample groups before expanding to larger-scale experimental rounds. Additional experiments examined ethnicity-associated demographic signals by modifying applicant names across multiple ethnicity groups, including Black, White, Hispanic, East Asian, and South Asian categories. The selected names were intended to act as demographic indicators while all other resume characteristics remained unchanged. This round was done twice, once with female-associated names and once with male-associated names. By separating resumes into controlled demographic groups, the study aimed to compare scoring behavior across cat-

egories while minimizing unrelated differences between applicants. This structure allowed for more direct observation of potential demographic patterns in LLM-based resume evaluation outputs.

2.3 Scoring Rubric

A standardized scoring rubric was developed to evaluate resumes consistently across all experimental rounds. The rubric divided resume evaluation into four primary categories: skills match, experience relevance, leadership and impact, and clarity and formatting. Each category contributed a fixed number of points toward a total score of 100. The skills match category evaluated the alignment between the skills of each applicant and the requirements listed in the job description. Experience relevance measured how closely prior work experience matched the target role. Leadership and impact assessed evidence of initiative, responsibility, and measurable contributions, while clarity and formatting evaluated resume organization, readability, and overall presentation quality. Using a fixed scoring rubric helped maintain consistency across experimental trials and reduced variation in evaluation criteria between different resume groups. The rubric was applied uniformly across all demographic categories and testing rounds to support more reliable comparison of scoring outcomes.

2.4 Experimental Rounds

This study was conducted across multiple experimental rounds designed to evaluate demographic patterns at different scales and points. Round 1: Gender-Based Pilot Testing/Early Round The initial experimental round focused on gender-associated demographic signals using a smaller pilot dataset. A total of 15 resumes were tested, consisting of 5 male-associated names, 5 female-associated names, and 5 gender-neutral names. All resumes contained identical qualifications, experience, formatting, and project descriptions. Three separate scoring rounds were conducted using the same job description and evaluation rubric in order to observe whether consistent scoring patterns emerged across demographic groups. Round 2: Large-Scale Gender Testing Following the pilot experiment, a larger-scale gender-based evaluation was conducted to expand the sample size and improve the reliability of observed trends. This round included approximately 300 resumes, consisting of 100 male-associated, 100 female-associated, and 100 gender-neutral resumes. The same scoring rubric, job description, and resume structure were maintained across all evaluations. Multiple scoring rounds were again performed to reduce the impact of random variation in model responses. Round 3: Ethnicity-Based Testing Additional experiments examined ethnicity-associated demographic signals using approximately 200 resumes distributed across multiple demographic groups. The tested categories included Black, White, Hispanic, East Asian, and South Asian name groups. As in previous rounds, all resumes remained identical except for demographic indicators such as applicant names. These experiments were designed to compare average scoring behavior across ethnicity-associated groups while maintaining controlled evaluation conditions. This round was done twice, once with male and once with female-associated names.

Results

3.1 Gender-Based Results

Gender Group	Run 1 Avg	Run 2 Avg	Run 3 Avg
Female	80.4	80.4	80.8
Male	78.6	79.2	78.4
Neutral	82.4	81.2	81

Figure 1: Round 1/Pilot Round, Gender-based Names

Gender Group	Run 1 Avg	Run 2 Avg	Run 3 Avg
Female	79.98	79.51	78.91
Male	80.69	80.59	80.34
Neutral	80.27	80.02	79.79

Figure 2: Round 2 Gender-based Names

3.2 Ethnicity-Based Results

Ethnicity Group	Average Score (Women)
Hispanic	78.6
East Asian	79.7
White	80.4
Black	79.3
South Asian	79.1

Figure 3: Female Names with Ethnicity Changes

Ethnicity Group	Average Score (Men)
White	81
South Asian	80.2
Hispanic	79.5
Black	79.6
East Asian	79.4

Figure 4: Male Names with Ethnicity Changes

Software System Development/Automation Framework

4.1 Motivation for Automation

As the scale of experimentation increased, manually processing and evaluating resumes became increasingly inefficient and difficult to manage consistently. To support larger-scale experimentation and improve reproducibility, an automated workflow system was developed using the n8n automation platform. The system was designed to automate resume retrieval, LLM-based evaluation, structured score extraction, and output collection for future large-scale testing.

4.2 Workflow Architecture

As the scale of experimentation increased, manually processing and evaluating resumes became increasingly inefficient and difficult to manage consistently. To support larger-scale experimentation and improve reproducibility, an automated workflow system was developed using the n8n automation platform. The system was designed to automate resume retrieval, LLM-based evaluation, structured score extraction, and output collection for future large-scale testing.

4.3 LLM Integration and Automated Scoring The automation framework incorporated OpenAI API-based resume evaluation to generate structured scoring outputs. Resume content and job descriptions were provided to the language model using standardized prompts aligned with the project scoring rubric. The model generated category-based evaluations including skills match, experience relevance, leadership and impact, and formatting quality. To improve consistency across evaluations, the workflow used fixed prompt structures and stan-

dardized output formatting. Structured JSON outputs were parsed automatically within the workflow before being exported into tabular result formats.

4.4 Output Processing and Scalability The workflow automatically collected scoring outputs and exported results into spreadsheet-based storage systems for future comparison and analysis. By automating repetitive processing tasks, the system improved scalability and reduced manual evaluation effort during experimentation. The automation framework was also designed to support future expansion, including larger resume datasets, additional demographic testing groups, and cross-model comparison experiments involving multiple LLM systems.

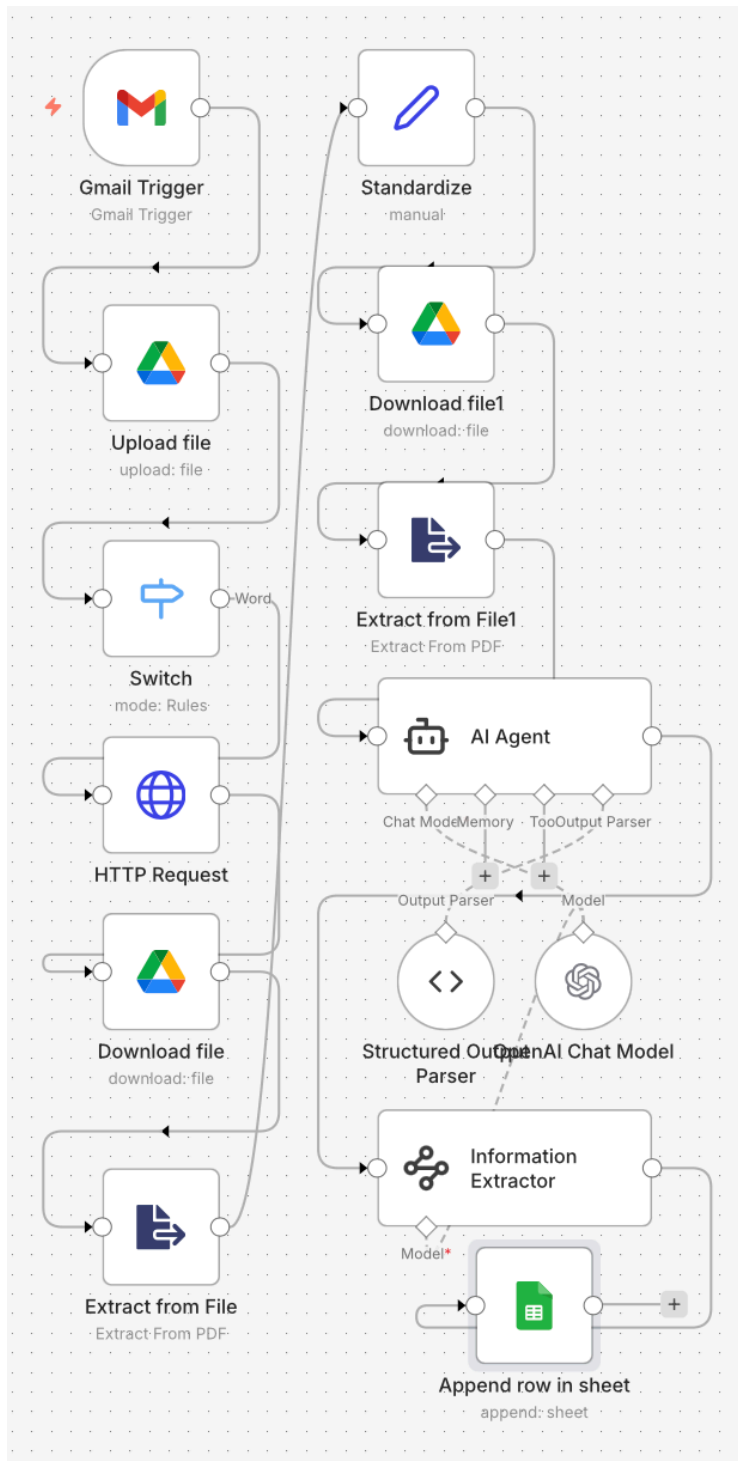


Figure 5: Workflow on n8n

Discussion

5.1 Interpretation of Results

The experimental results showed small but observable score differences across certain demographic groups during repeated testing rounds. Although the differences were generally limited in magnitude, some scoring patterns appeared consistently across multiple evaluations. These observations suggest that demographic signals such as names may influence the behavior of LLM-based resume screening systems under controlled conditions. At the same time, the results do not provide sufficient evidence to conclude that the evaluated systems exhibit definitive or systematic discrimination. Because score differences were relatively small and advanced statistical significance testing was not fully completed, the findings should be interpreted cautiously. Additional large-scale experimentation and formal statistical analysis would be necessary to determine whether the observed patterns reflect meaningful bias or normal variation in model outputs. The results also demonstrate the importance of controlled experimental design in fairness research. By maintaining identical resume content and modifying only demographic indicators, the study attempted to reduce unrelated variables that could otherwise affect scoring behavior. This approach helped isolate potential demographic sensitivity more directly than less controlled hiring datasets.

5.2 Ethical Implications

As AI-assisted hiring systems become more widely adopted, concerns regarding fairness and transparency continue to grow. Hiring decisions can significantly affect educational, financial, and career opportunities, making it important to evaluate whether automated systems treat applicants consistently across demographic groups. Even small differences in automated scoring behavior may become significant when applied across large applicant populations. The findings from this study highlight the importance of continued fairness testing in LLM-based hiring technologies. While the observed score differences were relatively small, the presence of consistent demographic patterns raises questions about how language models interpret demographic signals within resume evaluation tasks. These concerns become especially important when AI systems are used in high-impact decision-making environments with limited human oversight. This research also demonstrates the value of transparency and controlled evaluation methodologies in AI fairness studies. By systematically testing demographic variables under controlled conditions, researchers and developers may better identify potential weaknesses, unintended behaviors, and limitations within automated hiring systems. Continued evaluation and monitoring may help organizations develop more reliable and equitable AI-assisted recruitment practices.

5.3 Limitations

Several limitations were present within the scope of this study. Although the experiments used controlled resume content to isolate demographic variables, the

overall sample sizes remained limited compared to real-world hiring datasets. In addition, the study focused on a single resume template and a fixed job description, which may not fully represent the complexity and diversity of actual hiring scenarios. The study also primarily examined a single LLM-based evaluation framework rather than comparing multiple commercial hiring systems or language models. As a result, the findings should not be generalized to all AI-assisted hiring technologies. Future work involving larger datasets, additional models, and more advanced statistical testing would help strengthen the reliability and scope of the conclusions. Due to a lot of commercial systems and software not allowing people or non-businesses to purchase their service, it was not possible to conduct tests using them.

Conclusion and Future Work

This study explored whether demographic signals associated with gender and ethnicity influence the scoring behavior of LLM-based resume screening systems. Using a controlled experimental design, resumes with identical qualifications and experience were evaluated while demographic indicators such as names were systematically modified across multiple testing groups. The results showed small but observable scoring differences across certain demographic categories during repeated evaluations. Although the findings do not provide sufficient evidence to conclude the presence of definitive systematic bias, the experiments suggest that demographic signals may influence LLM-based resume evaluation behavior under some conditions. The study also demonstrated the value of controlled fairness testing methodologies for examining AI-assisted hiring systems. In addition to the experimental research, this project contributed an automated workflow framework designed to support scalable future experimentation and reproducible resume evaluation testing. The automation system enables more efficient large-scale analysis and may support future cross-model comparison studies. Future work may include larger datasets, additional demographic testing variables, more advanced statistical analysis, and evaluation across multiple language models and commercial hiring systems. Further research may also explore ranking-based selection methods, socioeconomic indicators, and broader fairness evaluation techniques for AI-assisted hiring technologies.

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